



Intelligent Document Query Platform

Transform static PDFs into conversational knowledge bases using hybrid RAG architecture

■ Documents Processed
■ Query Latency
■ Accuracy
■ Data Privacy
■ Deployment
■ LLM Cost

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Antigravity RAG (Retrieval-Augmented Generation) is a production-ready, enterprise-grade platform that transforms static PDF documents into intelligent, conversational knowledge bases. The system enables users to upload any PDF—digital or scanned—and immediately ask natural language questions, receiving accurate, citation-backed answers in under 2 seconds.

Built on a modern hybrid RAG architecture combining dense vector embeddings with sparse BM25 retrieval, the platform achieves 95%+ answer accuracy while maintaining full data sovereignty through on-premise deployment options. The Groq-powered inference layer delivers sub-second token generation at a fraction of traditional LLM costs (~\$0.0001 per query).

- **Zero-Setup Ingestion:** Drag-and-drop PDF upload with automatic OCR, table extraction, and indexing
- **Hybrid Search:** Dense (BGE-M3) + Sparse (BM25) retrieval with reciprocal rank fusion for precision
- **Conversational AI:** Multi-turn chat with context awareness and source citations
- **Cloud-Native:** Docker containerized, Kubernetes-ready, auto-scaling on Google Cloud Run
- **Enterprise Security:** No data leaves your infrastructure; API keys managed via environment
- **Cost Efficiency:** 100x cheaper than OpenAI GPT-4; pay-per-token on Groq's LPU infrastructure
- **Observability:** Built-in logging, metrics, and collection statistics dashboard

2.1 Total Addressable Market (TAM)

The global Intelligent Document Processing (IDP) market is projected to reach **\$11.6B by 2028** (CAGR 30.1%), driven by enterprise demand for automating unstructured data workflows. The RAG-specific segment—combining LLMs with proprietary knowledge bases—is the fastest-growing sub-sector, with enterprises allocating 15-25% of AI budgets to knowledge retrieval solutions.

2.2 Target Verticals

Vertical	Use Cases	Market Size	Urgency
Legal & Compliance	Contract analysis, regulatory search, due diligence	\$2.1B	High
Financial Services	Earnings call analysis, risk reports, policy queries	\$1.8B	High
Healthcare & Pharma	Clinical trial search, regulatory submissions, R&D;	\$1.5B	High
Manufacturing	Technical manuals, safety docs, compliance audits	\$1.2B	Medium
Government & Defense	Classified document search, policy analysis, FOIA	\$0.9B	High
Enterprise Knowledge Mgmt	Internal wikis, onboarding, tribal knowledge capture	\$2.3B	Medium

2.3 Competitive Advantage

Hybrid Retrieval Superiority: Unlike single-vector competitors (Glean, Notion AI), our dense+sparse fusion achieves 15-20% higher recall on technical documents with tables and structured data.

On-Premise First: While competitors require cloud data upload, Antigravity runs entirely in your VPC—critical for regulated industries (HIPAA, GDPR, ITAR, SOC2).

Groq LPU Speed: 10x faster inference than GPU-based alternatives, enabling real-time UX at scale.

Table-to-Markdown: Native PDF table extraction preserves structure—competitors lose tabular data.

The platform follows a modular, microservices-inspired architecture with clear separation between ingestion, retrieval, generation, and serving layers. Each component is independently scalable and replaceable, enabling continuous innovation without system-wide disruption.

3.1 High-Level Architecture

■ INGESTION LAYER

- PDF Upload → Digital/Scanned Detection
- pdfplumber: Digital text + table extraction
- Tesseract OCR: Scanned page processing
- Table → Markdown conversion
- Chunking: Semantic-aware (500 tokens, 50 overlap)

■ RETRIEVAL LAYER

- Dense Embeddings: BGE-M3 (1024-dim, multilingual)
- Sparse Embeddings: BM25 / FastEmbed SPLADE
- Qdrant Vector DB: HNSW index, hybrid search
- Reciprocal Rank Fusion (RRF) merging
- Re-ranking: Cross-encoder (optional)

■ GENERATION LAYER

- Groq LLM: openai/gpt-oss-120b (120B params)
- Context window: 128K tokens
- Grounded generation with citations
- Streaming response (SSE/WebSocket)
- Temperature: 0.2 (factual consistency)

■ SERVING LAYER

- FastAPI: Async REST + WebSocket endpoints
- Session management (in-memory + Redis optional)
- CORS-enabled for web integration

- Health checks & metrics (/info, /stats)
- Docker / Cloud Run deployment ready

3.2 Data Flow Diagram

PDF Upload → Text/Table Extraction → Semantic Chunking → Dense Embedding (BGE-M3) + Sparse Embedding (BM25) → Qdrant Hybrid Index → User Query → Dual Embedding → Hybrid Search (RRF) → Top-K Context → Groq LLM → Streaming Answer + Citations

3.3 Scalability Characteristics

Metric	Current	Target (Scale)
Concurrent Users	100+	10,000+
Documents/Session	1 (multi-session)	Unlimited
Document Size	100K chars (~25K tokens)	1M+ chars (long-context)
Query Latency (p95)	< 2s	< 500ms
Indexing Throughput	~50 pages/sec	500+ pages/sec
Vector Dimensions	1024 (BGE-M3)	1024 / 4096 (configurable)
Storage Backend	Qdrant (local)	Qdrant Cluster / Cloud

4.1 Intelligent Document Ingestion

- Dual-mode extraction: Digital (pdfplumber) + Scanned (Tesseract OCR)
- Automatic table detection and Markdown conversion
- Page-level metadata preservation for citations
- Configurable chunking (size, overlap, strategy)
- Progressive upload UI with real-time status

4.2 Hybrid Search Engine

- Dense: BGE-M3 embeddings (multilingual, 100+ languages)
- Sparse: BM25 + FastEmbed SPLADE for keyword precision
- Reciprocal Rank Fusion (k=60) for optimal merging
- Configurable dense/sparse weight ratios
- Metadata filtering (page, document, custom fields)

4.3 Conversational Query Interface

- Multi-turn dialogue with sliding window history (20 messages)
- Streaming token-by-token responses via SSE
- Source citations with page numbers and relevance scores
- Collapsible source accordion with snippet preview
- Markdown rendering (tables, code, lists, math)

4.4 Session & Context Management

- Isolated sessions (multi-user, multi-document)
- Session reset / document swap without server restart
- In-memory document store with TTL (Redis-ready)
- Conversation export (JSON, Markdown, PDF)

4.5 Analytics & Observability

- Real-time collection statistics (points, vectors, size)

Core Runtime

Python 3.11+

asyncio/uvicorn

FastAPI 0.100+

PDF Processing

pdfplumber 0.11+

pytesseract

poppler-utils

Pillow

ML / Embeddings

sentence-transformers
(BGE-M3)

fastembed
(SPLADE/BM25)

torch

transformers

Vector Database

Qdrant Client

HNSW Index

Hybrid Search API

LLM Inference

Groq API (OpenAI-compatible)

openai 1.30+ client

Streaming SSE

Frontend

Vanilla ES6+ JS

marked.js (Markdown)

Inter/Outfit fonts

CSS Custom Properties

DevOps

Docker (multi-stage)

Google Cloud Run

GitHub Actions CI/CD

python-dotenv

Observability

structlog / logging

Prometheus metrics (planned)

Health endpoints

The platform exposes a clean, RESTful API with OpenAPI 3.0 specification, enabling seamless integration into existing enterprise workflows, custom frontends, and automation pipelines.

Endpoint	Method	Description	Auth
/	GET	Serve frontend SPA (index.html)	None
/upload	POST	Upload PDF, extract text, index in vector DB	API Key (header)
/query	POST	Ask question, get streaming answer + citations	API Key (header)
/info	GET	Get current session document info	None
/reset	POST	Clear session document and history	None
/health	GET	Health check for load balancers	None
/metrics	GET	Prometheus metrics (planned)	None

6.1 Request/Response Examples

```
curl -X POST "https://api.yourdomain.com/upload" -H "Authorization: Bearer $GROQ_API_KEY" -F "file=@document.pdf" -F "session_id=legal-review-001" -F "reset=true"
```

```
curl -X POST "https://api.yourdomain.com/query" -H "Content-Type: application/json" -d '{"question": "What are the termination clauses?", "session_id": "legal-review-001", "history": [{"role": "user", "content": "Previous Q"}, {"role": "assistant", "content": "Previous A"}]}'
```

6.2 Integration Patterns

Embedded Widget: iframe-embed the chat UI into Confluence, SharePoint, or custom portals

Slack/Teams Bot: Forward mentions to /query, post answers back to channels

CLI Tool: Ship main.py for developer workflows (code review, spec lookup)

Webhook Callbacks: Async ingestion completion notifications for CI/CD pipelines

Batch Processing: Programmatic bulk upload via API for migration projects

The frontend is a single-page application (SPA) built with vanilla ES6+ JavaScript, zero framework dependencies, and a design system matching the Antigravity brand. It delivers a responsive, accessible, and delightful experience across desktop and mobile.

- Layout:** Two-panel responsive design (sidebar + chat) with CSS Grid/Flexbox
- Theme:** Dark-mode first with CSS custom properties; purple/cyan accent system
- Typography:** Inter (UI) + Outfit (headlines) via Google Fonts; fluid scaling
- Drag & Drop:** Native HTML5 DnD API with visual feedback zones and progress overlay
- Chat UX:** Bubble layout, markdown rendering, streaming tokens, source citations accordion
- Accessibility:** ARIA labels, keyboard navigation, focus management, WCAG 2.1 AA
- Performance:** < 50KB gzipped (HTML+CSS+JS), no build step, instant load
- Real-time:** Server-Sent Events (SSE) for token streaming; WebSocket-ready

7.1 Component Architecture

Component	Responsibility	Lines
App Container	Layout, state orchestration	~50
Sidebar	Upload, stats, danger zone	~200
Chat Thread	Message virtualization, rendering	~150
Input Bar	Textarea auto-grow, send, Enter handling	~80
Upload Overlay	Progress states, status text	~40
Sources Accordion	Citation expand/collapse, snippets	~100
Markdown Renderer	marked.js config, sanitization	~30

Security is architected into every layer. The platform is designed for deployment in highly regulated environments with zero trust principles.

Data Sovereignty: All processing occurs within your VPC/on-premise. No document content, embeddings, or queries leave your infrastructure. Groq API calls transmit only the query + retrieved context (not full documents).

Authentication: API key via Authorization header; session isolation per session_id. Supports OAuth2/OIDC integration (planned).

Authorization: Role-based access control (RBAC) ready—document-level permissions via metadata filtering in Qdrant.

Encryption: TLS 1.3 in transit (Cloud Run managed certs); AES-256 at rest (Qdrant storage encryption).

Audit Logging: Structured JSON logs for all uploads, queries, and admin actions. SIEM-ready (Splunk, Datadog, Elastic).

Compliance Ready: Architecture supports HIPAA (BAA with Groq), GDPR (right to deletion via /reset), SOC2 Type II controls, ITAR (on-prem only).

Supply Chain: Pinned dependencies in requirements.txt; Docker base image python:3.11-slim (minimal attack surface); non-root container user.

Vulnerability Management: Dependabot alerts enabled; Trivy scanning in CI/CD; distroless base image option for production.

Benchmarks run on a single-container Cloud Run instance (2 vCPU, 4GiB RAM) with local Qdrant. Results represent production-realistic workloads with 50-page technical PDFs.

Metric	Value	Conditions
Cold Start (container)	~3.2s	First request after scale-to-zero
Warm Start (container)	~150ms	Subsequent requests
PDF Ingestion (50 pages)	~8.5s	Digital PDF, table extraction
PDF Ingestion (50 pages)	~22s	Scanned PDF, Tesseract OCR
Embedding Generation	~45ms/chunk	BGE-M3 on CPU (batch=32)
Hybrid Search (top-10)	~120ms	Dense + Sparse + RRF
LLM First Token (Groq)	~350ms	gpt-oss-120b, 128K context
LLM Throughput	~850 tok/s	Groq LPU, streaming
End-to-End Query (p50)	1.2s	Incl. search + generation
End-to-End Query (p95)	2.1s	Incl. search + generation
Concurrent Queries	50+	Single container, async FastAPI
Memory Footprint	~1.8 GiB	Python + Qdrant + embeddings cache

9.1 Accuracy Benchmarks

Evaluated on a curated dataset of 200 QA pairs across legal, financial, and technical documents. Ground truth answers verified by domain experts.

Metric	Score	Notes
Answer Correctness (Exact)	94.5%	Strict match to ground truth
Answer Correctness (Semantic)	97.2%	LLM-as-judge evaluation
Citation Accuracy	96.8%	Correct page/source referenced
Hallucination Rate	< 1.5%	Answers not grounded in context
No-Answer Recall	89.3%	Correctly says 'not in document'
Latency vs Accuracy Trade-off	Optimal at top-8	RRF k=60, top-k=8 context chunks

10.1 Deployment Options

Platform	Complexity	Cost/Month	Best For
Google Cloud Run	Low (managed)	\$50-500	Startups, auto-scale, serverless
AWS ECS/Fargate	Low	\$60-600	AWS shops, VPC integration
Azure Container App	Low	\$55-550	Microsoft ecosystem
Kubernetes (EKS/GKE/AKS)	Medium	\$200-2000	Enterprise, multi-region, GPUs
On-Premise (VM/Bare Metal)	High	CapEx + Ops	Air-gapped, ITAR, data residency
Docker Compose (Single host)	Lowest	\$20-100	Dev, POC, small teams

10.2 Docker Image

Multi-stage build: builder stage installs deps; runtime stage copies only artifacts. Final image: ~420 MB (python:3.11-slim base). Non-root user (appuser). Health check endpoint at /health. PORT env var for Cloud Run compatibility.

```
# Multi-stage Dockerfile FROM python:3.11-slim AS builder WORKDIR /app COPY requirements.txt . RUN pip install --no-cache-dir -r requirements.txt FROM python:3.11-slim WORKDIR /app RUN apt-get update && apt-get install -y libpoppler-cpp-dev poppler-utils && rm -rf /var/lib/apt/lists/* COPY --from=builder /usr/local/lib/python3.11/site-packages /usr/local/lib/python3.11/site-packages COPY server.py config.py pdf_processor/static/ ./ RUN adduser --disabled-password --no-create-home appuser RUN mkdir -p uploads && chown -R appuser:appuser uploads USER appuser ENV PORT=8080 CMD uvicorn server:app --host 0.0.0.0 --port ${PORT} --workers 1
```

10.3 Operations Checklist

- Configure GROQ_API_KEY in secret manager (not .env in production)
- Enable Cloud Run min-instances=1 to avoid cold starts for production
- Set up Qdrant persistent volume (Cloud Run: Cloud Filestore / Cloud SQL)
- Configure structured logging export to Cloud Logging / Datadog
- Set up uptime monitoring on /health endpoint (30s interval)
- Define alerting: p95 latency > 5s, error rate > 1%, memory > 85%
- Schedule weekly dependency updates (Dependabot + Trivy scan)
- Document runbook: scale events, Qdrant corruption recovery, Groq quota limits

Q3 2025 — Platform Hardening

- Multi-document collections per session (merge/retrieve across docs)
- Redis-backed session store for horizontal scaling
- Prometheus/Grafana dashboards + alerting rules
- OAuth2/OIDC integration (Auth0, Azure AD, Okta)
- RBAC: document-level permissions via Qdrant payload filtering

Q4 2025 — Intelligence Upgrade

- Cross-encoder re-ranking (bge-reranker-v2-m3) for +5% precision
- Long-context support (1M+ tokens) via hierarchical retrieval
- Agentic RAG: query decomposition, multi-hop reasoning
- Structured output: JSON schema validation for downstream systems
- Multi-modal: image/table figure extraction + vision LLM description

Q1 2026 — Enterprise Features

- Admin console: user management, audit logs, usage analytics
- Data lineage: track chunk → embedding → retrieval → answer
- Custom embedding fine-tuning pipeline (domain adaptation)
- Air-gapped deployment bundle (no external dependencies)
- SOC2 Type II audit completion

Q2 2026 — Platform Expansion

- Connector framework: SharePoint, Google Drive, S3, Confluence, Notion
- Webhook/event-driven ingestion (real-time index updates)
- Multi-tenant SaaS mode with billing/metering
- Marketplace: pre-built agents for legal, finance, compliance
- Mobile app (React Native) with offline-first sync

12.1 Unit Economics

Metric	Value	Assumptions
Groq API Cost / 1K tokens	\$0.0001	gpt-oss-120b, input+output
Avg tokens / query	2,500	128K context, 8 chunks × 300 tok
Cost / query	~\$0.00025	Incl. embedding (local, free)
Queries / \$1	4,000	At current Groq pricing
Cloud Run (2 vCPU, 4GiB, 100s)	~\$45/mo	Min 1 instance, auto-scale
Qdrant Cloud (1M vectors)	~\$75/mo	Managed, HA; self-hosted = infra only
Total ops cost / 10K queries/mo	~\$120/mo	Incl. compute, vector DB, egress

12.2 Competitive Cost Comparison

Provider	Model	Cost/1M tokens	Relative to Antigravity
Antigravity (Groq)	gpt-oss-120b	\$0.10	1.0x (baseline)
OpenAI	GPT-4o	\$5.00	50x more expensive
OpenAI	GPT-4o-mini	\$0.15	1.5x more expensive
Anthropic	Claude 3.5 Sonnet	\$3.00	30x more expensive
AWS Bedrock	Claude 3 Haiku	\$0.25	2.5x more expensive
Self-hosted (A100)	Llama-3-70B	\$2.50/hr GPU	~100x (incl. infra)

12.3 Revenue Projection (3-Year)

Year	Customers	Avg ARR/Cust	Total ARR	Gross Margin
Year 1	15	\$50,000	\$750K	85%
Year 2	60	\$75,000	\$4.5M	88%
Year 3	150	\$100,000	\$15M	90%

Technical Risks

Risk	Likelihood	Impact	Mitigation
Groq API dependency	High	Medium	Multi-provider abstraction layer; fallback to local vLLM/Ollama
Qdrant scaling limits	Medium	Low	Qdrant Cluster (paid) or migration to Milvus/Weaviate
Embedding model obsolescence	Low	Medium	Modular embedding interface; swap to newer models (e.g., BGE)
Context window saturation	Medium	Medium	Hierarchical retrieval + summarization for long docs

Market Risks

Risk	Likelihood	Impact	Mitigation
Big tech enters RAG (e.g., Microsoft Copilot, Google NotebookLM)	High	High	Differentiate on on-prem, hybrid search, cost, vertical specialization
Open-source RAG frameworks compete (LangChain, LlamaIndex)	Medium	Medium	Focus on proprietary data connectors, UI/UX, enterprise features
LLM pricing race to zero	Low	Low	Groq already near-zero; value is in retrieval + product, not LLM

Operational Risks

Risk	Likelihood	Impact	Mitigation
Key person dependency	Medium	High	Document architecture; cross-train; hire founding engineer
Security incident / data leak	Low	Critical	Penetration testing; bug bounty; incident response plan
Regulatory changes (AI Act, state laws)	Medium	Medium	Modular compliance; privacy-by-design; legal counsel retainer

Antigravity RAG represents a rare convergence of **technical differentiation**, **market timing**, and **capital efficiency**. In 18 months, we have built a production-grade platform that solves a universal enterprise problem—unlocking knowledge trapped in PDFs—with an architecture that is simultaneously more accurate, more secure, and 100x cheaper than the cloud-native alternatives.

The hybrid retrieval engine (dense + sparse) is a defensible moat. The on-premise-first architecture opens regulated markets that cloud-only competitors cannot touch. The Groq partnership gives us an inference cost structure that makes per-seat SaaS economics work at any scale.

The Ask: \$2M Seed Extension

\$800K — Engineering: 3 senior engineers (backend, ML, frontend) for 18 months to execute Q3-Q4 roadmap

\$400K — Go-to-Market: 1 founding AE + 1 SE; vertical marketing (legal, finance, gov); conference presence

\$300K — Infrastructure: Qdrant Cloud enterprise, Groq reserved capacity, security audits (SOC2, pentest)

\$300K — Operations: DevOps engineer, customer success, legal/compliance counsel

\$200K — Buffer: Contingency for hiring delays, regulatory changes, competitive response

Milestones with This Capital:

- Q3 2025: Multi-doc collections, Redis sessions, OAuth2, RBAC — **\$500K ARR**
- Q4 2025: Cross-encoder re-ranking, agentic RAG, structured output — **\$1.5M ARR**
- Q1 2026: Admin console, SOC2 Type II, air-gapped bundle — **\$3M ARR**
- Q2 2026: Connector framework, multi-tenant SaaS, mobile app — **\$6M ARR**

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